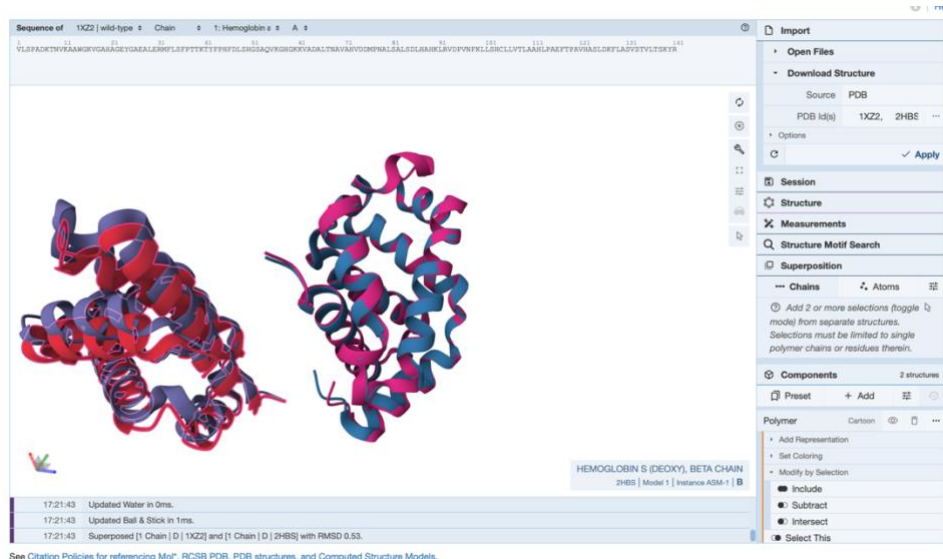
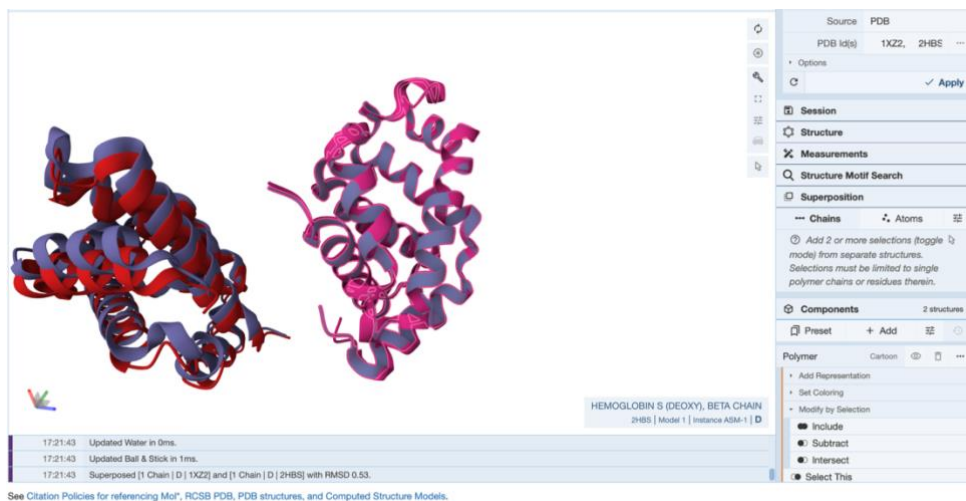


Comparing 1XZ2 (wild type) with 2HBS (mutated: beta Glu6 -> Val)

Superimposing all subunits



Superimposing only beta subunits



ClinVar Table (Germline Mutations)

Disease	Description	Gene	Example Variant	Mutation type	Functional effect
Hemochromatosis	Excess iron	HJV	NM_213653.3:c.959G>T	Missense	Likely LOF
Thalassemia	tools predict a damaging effect of the variant on protein function.	ALPL	NM_000478.6(ALPL):c.659G>A (p.Gly220Glu)	Missense	Likely pathogenic
Cystic Fibrosis		C1QB	NM_001378156.1(C1QB):c.490G>A (p.Gly164Ser)	Missense	Uncertain significance
Tay-Sachs disease		GM2A	NM_000405.5(GM2A):c.33C>T (p.Ile11=)	synonymous	Benign/Likely benign

ClinVar Somatic Mutation

NRAS, NM_002524.5(NRAS):c.182A>G (p.Gln61Arg), Missense